

Transformer Memory and Compute Accounting Summary

Systems Engineering Reference Sheet

1 Core Architectural Variables

Before calculating system requirements, we define the standard hyperparameter variables used in transformer scaling laws:

- b : Batch size
- T : Sequence length (Context window)
- V : Vocabulary size
- D : Hidden dimension (d_{model})
- L : Number of transformer layers
- h : Number of attention heads
- d_k, d_v : Dimensions of queries/keys and values (D/h)
- D_{ff} : Inner feed-forward network dimension

2 Memory Accounting (VRAM)

Memory is tracked in four distinct buckets during training. The final peak memory dictates the maximum possible batch size a GPU can support before triggering an Out-of-Memory (OOM) error.

2.1 1. Parameters (P)

The total count of learnable weights in the model.

$$\text{Outer Layers: } 2VD + D$$

$$\text{One Transformer Block: } 2D + 2hD(d_k + d_v) + 3D_{ff}D$$

$$\text{TotalParameters(P) : } 2VD + D + L(2D + 2hD(d_k + d_v) + 3D_{ff}D)$$

2.2 2. Activations (A)

The intermediate forward-pass tensors that must be saved in memory to compute the chain rule during the backward pass.

$$\text{Outer Layers: } 2bTD + 2bTV$$

$$\text{One Transformer Block: } 4bTD + 2bTh(d_k + d_v) + 2bhT^2 + 4bTd_{ff}$$

$$\text{TotalActivations(A) : } 2bTD + 2bTV + L(4bTD + 2bTh(d_k + d_v) + 2bhT^2 + 4bTd_{ff})$$

2.3 3. Gradients and Optimizer State

- **Gradients:** One tensor per parameter. Size = P .
- **AdamW Optimizer:** Requires tracking the first moment (m) and second moment (v) for every parameter. Size = $2P$.

2.4 Total Peak Memory Formula

By summing Activations (A), Parameters (P), Gradients (P), and AdamW State (2P), we arrive at the master equation for memory footprint in floats:

$$\text{Peak Memory (floats)} = \mathbf{A} + \mathbf{4P} \quad (1)$$

Note: Multiply this total by 4 to get the physical footprint in bytes (assuming standard FP32 precision).

3 Compute Accounting (FLOPs)

Compute measures the mathematical operations (Floating Point Operations) required by the processing cores, completely separate from memory storage.

3.1 The 6PTb Rule

This approximation captures the massive matrix multiplications during training. Because one Multiply-Accumulate (MAC) operation equals 2 FLOPs:

- **Forward Pass:** $2 \cdot P \cdot T \cdot b$
- **Backward Pass:** $4 \cdot P \cdot T \cdot b$ (Double the forward pass due to ∇X and ∇W)
- **Total Math per Step:** 6PTb

If calculating total compute for an entire training run based on a target dataset size, the equation simplifies to: $6 \cdot P \cdot N_{\text{tokens}}$.

3.2 AdamW Optimizer Step

The optimizer relies strictly on element-wise operations and scales only with the parameter count, completely independent of context length or batch size.

$$\text{Optimizer FLOPs per Step} = 14 \cdot P \quad (2)$$

4 Hardware and Throughput Estimation

To calculate how fast a model will train on a specific chip, we divide the total mathematical workload by the chip's realistic processing speed.

$$\text{Estimated Time} = \frac{\text{Total Training FLOPs}}{\text{Hardware Peak FLOPs/sec} \times \text{Efficiency \%}} \quad (3)$$

Note on Efficiency: Frameworks like PyTorch on Apple's MPS backend typically operate at roughly 14% efficiency of the theoretical Peak FP32 limit due to memory bandwidth bottlenecks and kernel overhead.

5 Instantiated Examples

5.1 Example 1: GPT-2 XL

$$V = 50257, T = 1024, L = 48, D = 1600, h = 25, d_{ff} = 4288$$

- **Total Parameters (P):** 1,640,452,800
- **Peak Memory Bytes:** $16,379,944,960 \times b + 26,247,244,800$
- **Max Batch Size on 80GB GPU:** $b = 3$
- **Optimizer Compute ($14P$):** ~ 23 GigaFLOPs

5.2 Example 2: Custom Assignment Spec (Base M5 Hardware)

$$V = 10000, T = 256, L = 4, D = 512, h = 16, d_{ff} = 1344$$

- **Total Parameters (P):** 22,696,448 (~ 17 M non-embedding)
- **Peak Memory Bytes:** $93,880,320 \times b + 363,143,168$
- **VRAM at $b = 32$:** ~ 3.37 GB (Easily fits 16GB Unified Memory)
- **Total Compute (40.96M tokens):** ~ 5.58 PetaFLOPs
- **Estimated Run Time (Base M5):** ~ 2 hours and 38 minutes